

Theoretical Framework: Entanglement Structure of the Lattice Schwinger Model

Section 05 — Entanglement Structure & Quantum Information

Repository: `lattice-to-lindblad/05_Entanglement_Structure_QI/`

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1. Motivation & Physical Setup

1.1 What physical system are we studying?

We study the **lattice Schwinger model** — quantum electrodynamics in one spatial dimension, discretized on a staggered lattice of N sites with lattice spacing a . The model describes a $U(1)$ gauge theory coupled to a single flavor of massive fermion, and is the simplest gauge theory admitting confinement, chiral symmetry breaking, and a nontrivial vacuum structure.

We assume open boundary conditions (OBC) throughout this section. This is the natural choice for DMRG-based MPS studies because OBC avoid the entanglement overhead of periodic boundaries and keep the MPS bond dimension finite at fixed accuracy. All ground-state entanglement observables are defined with respect to this convention.

After the Jordan–Wigner transformation and gauge-field elimination via Gauss’s law, the working Hilbert space is represented by N spin-1/2 degrees of freedom, while the electric field is reconstructed from the matter configuration. The Hamiltonian inherited from the base framework is

$$H_{\text{Schwinger}} = w \sum_{n=0}^{N-2} (\sigma_n^+ \sigma_{n+1}^- + \sigma_n^- \sigma_{n+1}^+) + \frac{\mu}{2} \sum_{n=0}^{N-1} (-1)^n \sigma_n^z + J \sum_{n=0}^{N-2} L_n^2, \quad (1.1)$$

with reconstructed electric field

$$L_n = \ell_0 + \frac{1}{2} \sum_{k=0}^n (\sigma_k^z + (-1)^k). \quad (1.2)$$

In the present Section 05 analysis we remain in the zero-background sector

$$\ell_0 = 0. \quad (1.3)$$

The associated staggered charge operator is

$$\rho_n = \frac{1}{2} (\sigma_n^z + (-1)^n), \quad (1.4)$$

so that $\rho_n \in \{0, 1\}$ on even sites and $\rho_n \in \{-1, 0\}$ on odd sites. This convention is inherited from the integrated-out Hamiltonian in the base framework and must remain synchronized with the implementation in `04_Continuum_Physics_Results/schwinger_dmrg.py`.

1.2 Why does this system matter for quantum information?

The Schwinger model is a canonical testbed for quantum simulation of gauge theories, but its **entanglement structure** is itself a mathematically meaningful object. Gauss’s law correlates the two halves of a bipartition through the cut flux, so the reduced density matrix acquires a nontrivial block structure in charge / flux sectors. Consequently, the entanglement entropy splits into an inter-sector Shannon contribution and weighted intra-sector entropies, the entanglement spectrum

is organized into symmetry-resolved branches, and tensor-network compressibility is enhanced whenever only a small number of sectors carry substantial Schmidt weight.

The present extension adds two Stage 3 upgrades to this core story:

- `schwinger_symmetry_resolved_entanglement.py`, which asks how the bipartite entanglement is distributed across constrained sectors; and
- `open_schwinger_entanglement_dynamics.py`, which asks how weak openness modifies entanglement growth and tensor-network compressibility during Schwinger dynamics.

These scripts are framed as **** upgrade scripts****, not replacements for the core entropy / spectrum / Schmidt-decay pipeline.

1.3 Why these computational methods?

We use **DMRG** to obtain ground-state MPS representations because the zero-background Schwinger model in the confining regime is expected to be efficiently approximated by low-entanglement tensor networks. The existing TeNPy-based pipeline in Section 04 already produces accurate MPS ground states at modest bond dimension. Section 05 adds only the information-theoretic layer that reads Schmidt data, reduced-density structure, and sector labels from that MPS.

For the open-system extension, the relevant tensor-network object may be either a pure-state MPS (closed evolution or quantum-trajectory evolution) or an MPO / purified-state representation of a density matrix. The framework below distinguishes these cases explicitly, because the quantity called “entanglement entropy” in a closed pure state is not identical to the subsystem entropy of a mixed state, nor to the operator-space entanglement of an MPO.

2. Schwinger-Model Hamiltonian and Hilbert-Space Structure

2.1 Gauss-law-constrained Hilbert space

The working spin Hilbert space has dimension 2^N . After gauge elimination, the electric field is no longer independent; it is reconstructed from the spin configuration via Eq. (1.2). The local Gauss-law relation reads

$$L_n - L_{n-1} = \rho_n = \frac{1}{2}(\sigma_n^z + (-1)^n). \quad (2.1)$$

Given OBC and fixed ℓ_0 , each spin configuration determines a unique electric-field profile. Thus gauge elimination does not further reduce the dimension of the spin working space, but it does constrain the allowed combinations of bipartite quantum numbers across a cut:

$$\dim \mathcal{H}_{\text{work}} = 2^N. \quad (2.2)$$

2.2 Bipartition and the cut electric flux

For a bipartition at bond ℓ , define

$$A = \{0, 1, \dots, \ell\}, \quad B = \{\ell + 1, \dots, N - 1\}. \quad (2.3)$$

The ground state admits a Schmidt decomposition

$$|\Psi_0\rangle = \sum_{\alpha=1}^{\chi_{\text{exact}}} \lambda_{\alpha} |L_{\alpha}\rangle_A \otimes |R_{\alpha}\rangle_B, \quad (2.4)$$

with $\lambda_{\alpha} \geq 0$ and $\sum_{\alpha} \lambda_{\alpha}^2 = 1$.

The total staggered charge is

$$Q = \sum_{n=0}^{N-1} \rho_n = \frac{1}{2} \sum_{n=0}^{N-1} (\sigma_n^z + (-1)^n). \quad (2.5)$$

If the state is chosen in a definite total-charge sector, each Schmidt pair carries compatible block charges

$$Q_A |L_{\alpha}\rangle = q_{\alpha} |L_{\alpha}\rangle, \quad Q_B |R_{\alpha}\rangle = (Q_{\text{tot}} - q_{\alpha}) |R_{\alpha}\rangle, \quad (2.6)$$

where

$$Q_A = \sum_{n \in A} \rho_n, \quad Q_B = \sum_{n \in B} \rho_n. \quad (2.7)$$

The cut flux is therefore

$$L_\ell = \ell_0 + Q_A = q_\alpha \quad (\ell_0 = 0). \quad (2.8)$$

The Hilbert-space structure is thus a direct sum of sector subspaces, while the state itself is an ordinary sum of orthogonal sector components:

$$\mathcal{H}_A \otimes \mathcal{H}_B = \bigoplus_q \mathcal{H}_A^{(q)} \otimes \mathcal{H}_B^{(Q_{\text{tot}} - q)}, \quad (2.9)$$

$$|\Psi_0\rangle = \sum_q \sum_{\alpha \in q} \lambda_\alpha^{(q)} |L_\alpha^{(q)}\rangle_A \otimes |R_\alpha^{(q)}\rangle_B. \quad (2.10)$$

This relation is the mathematical starting point for both the existing spectrum script and the planned symmetry-resolved extension.

3. Entanglement Observables from First Principles

3.1 Von Neumann entropy across a bond

For a pure ground state, the reduced density matrix of the left block is

$$\rho_A = \text{Tr}_B |\Psi_0\rangle\langle\Psi_0| = \sum_\alpha \lambda_\alpha^2 |L_\alpha\rangle\langle L_\alpha|, \quad (3.1)$$

so the bond entropy is

$$S(\ell) = -\text{Tr}[\rho_A \ln \rho_A] = -\sum_\alpha \lambda_\alpha^2 \ln \lambda_\alpha^2. \quad (3.2)$$

In the MPS representation, the λ_α are exactly the singular values stored at the cut.

3.2 Sector decomposition of the entropy

Let the sector weights be

$$p_q(\ell) = \sum_{\alpha \in q} \lambda_\alpha^{(q)2}, \quad \sum_q p_q(\ell) = 1, \quad (3.3)$$

and define normalized intra-sector probabilities

$$p_{\alpha|q}(\ell) = \frac{\lambda_\alpha^{(q)2}}{p_q(\ell)}. \quad (3.4)$$

The full entropy decomposes as

$$S(\ell) = H_{\text{sector}}(\ell) + \sum_q p_q(\ell) S_q(\ell), \quad (3.5)$$

where

$$H_{\text{sector}}(\ell) = - \sum_q p_q(\ell) \ln p_q(\ell), \quad S_q(\ell) = - \sum_{\alpha \in q} p_{\alpha|q}(\ell) \ln p_{\alpha|q}(\ell). \quad (3.6)$$

Only the first term is the inter-sector Shannon contribution. The full entropy is not bounded solely by the number of sectors, because each sector may contain nontrivial internal entanglement.

3.3 Entanglement spectrum and branch structure

Define entanglement energies

$$\xi_\alpha = - \ln \lambda_\alpha^2. \quad (3.7)$$

The spectrum decomposes into symmetry-resolved branches

$$\{\xi_\alpha\} = \bigsqcup_q \{\xi_\alpha^{(q)}\}, \quad \xi_\alpha^{(q)} = - \ln \lambda_\alpha^{(q)2}. \quad (3.8)$$

If A contains $N_{\text{even}}^{(A)}$ even sites and $N_{\text{odd}}^{(A)}$ odd sites, then the allowed left-block charges satisfy

$$-N_{\text{odd}}^{(A)} \leq q_A \leq N_{\text{even}}^{(A)}. \quad (3.9)$$

Likewise the right block must satisfy the corresponding parity-dependent bounds. This proposition matters directly for the planned sector-weight plots, because a sector is physically admissible only if it is realizable on both sides of the cut.

3.4 Schmidt decay and effective compressibility

The ordered Schmidt values

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{\chi_{\text{exact}}} \geq 0 \quad (3.10)$$

quantify the compressibility of the state across the cut. The discarded weight after keeping the first χ' values is

$$\epsilon(\chi') = 1 - \sum_{\alpha=1}^{\chi'} \lambda_{\alpha}^2 = \sum_{\alpha=\chi'+1}^{\chi_{\text{exact}}} \lambda_{\alpha}^2. \quad (3.11)$$

Define also the cumulative retained weight

$$W(\chi') = \sum_{\alpha=1}^{\chi'} \lambda_{\alpha}^2, \quad \chi_{\text{eff}}(\epsilon) = \min\{\chi' : W(\chi') \geq 1 - \epsilon\}. \quad (3.12)$$

These quantities become central again in the open-system extension, where compressibility is tracked either through ordinary Schmidt values of a pure state or through operator-space Schmidt values of an MPO.

3.5 Mixed-state caution for the open-system extension

The dynamics script introduces a density matrix $\rho(t)$ obeying a Lindblad equation. In that setting, one must distinguish three diagnostics:

1. **Subsystem entropy of a mixed state,**

$$S_A(t) = -\text{Tr}[\rho_A(t) \ln \rho_A(t)], \quad \rho_A(t) = \text{Tr}_{\bar{A}} \rho(t), \quad (3.13)$$

which quantifies local uncertainty and correlations but is not, by itself, a pure-state entanglement monotone.

2. **Operator-space entanglement entropy (OSEE),** obtained from a Schmidt decomposition of the vectorized density operator $\text{vec}(\rho(t))$ across the cut,

$$\text{vec}(\rho(t)) = \sum_a s_a(t) \text{vec}(L_a(t)) \otimes \text{vec}(R_a(t)), \quad \sum_a s_a(t)^2 = 1, \quad (3.14)$$

$$S_{\text{op}}(\ell, t) = - \sum_a s_a(t)^2 \ln s_a(t)^2, \quad (3.15)$$

which directly measures MPO compressibility; and

3. **Trajectory-averaged pure-state entropy,** relevant only if the dynamics are implemented via quantum trajectories,

$$\bar{S}(\ell, t) = \frac{1}{R} \sum_{r=1}^R S_r(\ell, t). \quad (3.16)$$

This last quantity is unraveling-dependent and should be labeled as such. The preferred density-matrix-level diagnostic for tensor-network cost is Eq. (3.15).

4. Tensor-Network Representation and Compressibility

4.1 MPS canonical form

An MPS in mixed-canonical form centered at bond ℓ reads

$$|\Psi\rangle = \sum_{\alpha=1}^{\chi_\ell} \lambda_\alpha^{(\ell)} |L_\alpha^{(\ell)}\rangle \otimes |R_\alpha^{(\ell)}\rangle. \quad (4.1)$$

This is precisely the Schmidt decomposition across that bond, so all ground-state observables in Sections 3.1–3.4 are read directly from the canonicalized MPS.

4.2 Convergence and numerical reliability

The entanglement observables are reliable only insofar as the MPS or MPO calculation is converged. We therefore use an operational convergence criterion rather than an unsupported universal error law:

$$\delta S_{\chi \rightarrow \chi'}(\ell) = |S_{\chi}(\ell) - S_{\chi'}(\ell)|, \quad \chi' > \chi. \quad (4.2)$$

For the open-system extension, an analogous comparison should be performed for $S_A(t)$ or $S_{\text{op}}(t)$ at fixed times. The claim is then empirical stability under increasing bond dimension and time-step refinement, not an unproven asymptotic bound.

4.3 Reused infrastructure

```
schwinger_dmrg.py → saves converged MPS
                    ↓
utils_entanglement.py → loads MPS, extracts Schmidt data, resolves sectors
                    ↓
schwinger_entanglement_entropy.py
schwinger_entanglement_spectrum.py
schmidt_decay_analysis.py
```

Stage 3:

```
schwinger_symmetry_resolved_entanglement.py
open_schwinger_entanglement_dynamics.py
```

The core principle is to reuse trusted state-generation code. Neither extension should reimplement the DMRG ground-state solver from scratch.

5. Reference Paper Connections

5.1 Hastings and the area-law perspective

The area-law literature supplies the conceptual benchmark that low-entanglement one-dimensional states are efficiently represented by tensor networks. Because the integrated-out Hamiltonian contains long-range interactions, we use the area-law picture as a physical expectation and numerical benchmark rather than as a theorem applied verbatim to Eq. (1.1).

5.2 Calabrese–Cardy caveat

The present zero-background framework does **not** adopt a $c = 1$ conformal fit as a validation target. The relevant criticality claims depend on the background-field sector and must not be imported into the $\ell_0 = 0$ analysis without additional justification.

5.3 Li–Haldane and structural use of the entanglement spectrum

The entanglement spectrum is used here as a structural diagnostic rather than as a literal edge-theory proxy. This methodological lesson extends naturally to the symmetry-resolved script, whose purpose is to identify which sectors actually carry the Schmidt weight.

5.4 Gauge-theory tensor networks

The broader MPS literature on lattice gauge theories motivates the use of conserved virtual charges and sector-resolved bond data. In the present integrated-out formulation, those virtual charges are interpreted as reconstructed cut fluxes rather than as explicit link variables retained in the local Hilbert space.

6. Stage 3 Extensions

Stage 3 status. The two scripts in this section are upgrades on top of the core Section 05 project. They are not replacements for the entropy-profile / entanglement-spectrum / Schmidt-decay pipeline. The recommended workflow is to implement *one* fully and only then start the second.

6.1 Priority logic

The higher-theoretical-value extension for a symmetry / entanglement oriented application is `schwinger_symmetry_resolved_entanglement.py`. The higher-implementation-reliability extension, especially if an open-system evolution path already exists elsewhere in the repo, is `open_schwinger_entanglement_dynamics.py`. The practical rule is straightforward:

- if sector labels are already accessible from conserved MPS bond charges, implement the symmetry-resolved script first;
- if sector extraction is basis-dependent or awkward, switch early to the open-system dynamics script and finish that one cleanly.

6.2 Extension A — `schwinger_symmetry_resolved_entanglement.py`

6.2.1 Scientific purpose This script upgrades the project from “how much entanglement is present?” to the more structural question *which constrained sectors actually carry the entanglement across a bipartition?* The clean scientific statement is:

We analyze not only the amount of entanglement in the lattice Schwinger model, but how that entanglement is distributed across physically relevant constrained sectors.

This is the most natural extension of Eqs. (2.10) and (3.3)–(3.6), and it aligns directly with the charge / flux organization already implicit in the base framework.

6.2.2 Frozen minimum scope The first version should remain narrow: one Schwinger model family, ground state only, one or two representative parameter points, one representative cut (preferably the center cut), and one robustly computable sector label. The goal is **not** a full symmetry-resolved-entanglement paper.

6.2.3 Precise observables At minimum the script should compute, for one chosen cut ℓ :

1. Sector weights

$$p_q(\ell) = \sum_{\alpha \in q} \lambda_{\alpha}^{(q)2}, \quad (6.1)$$

which are the primary observable.

2. Cumulative sector concentration

$$P_{\leq k}(\ell) = \sum_{j=1}^k p_{q_j}(\ell), \quad (6.2)$$

where sectors are ordered either by descending weight or by sector label.

3. Sector-resolved entropy diagnostics, if straightforward:

$$h_q(\ell) = -p_q(\ell) \ln p_q(\ell), \quad c_q(\ell) = p_q(\ell) S_q(\ell), \quad (6.3)$$

so that

$$S(\ell) = \sum_q h_q(\ell) + \sum_q c_q(\ell). \quad (6.4)$$

If the full decomposition in Eq. (6.4) is implementation-heavy, sector weights alone are already scientifically meaningful and should be prioritized.

6.2.4 Choice of sector label The preferred sector label is the left-block conserved quantity naturally exposed by the symmetry-preserving MPS bond. In the present zero-background convention, the most physical interpretation is the cut charge / cut flux

$$q \equiv Q_A = L_\ell. \quad (6.5)$$

If the exact physical interpretation of the virtual bond charge is not fully transparent, the first implementation may still use a generic sector label provided that its relation to the TeNPy bond charge is explicitly documented and internally validated.

6.2.5 Deliverables and interface The planned outputs are:

- `symmetry_resolved_entanglement_schwinger.csv`,
- `symmetry_resolved_entanglement.png`,

- `run_metadata.json`.

The recommended figure is a two-panel layout: sector weights in the left panel and sector-resolved entropy contributions or cumulative weight in the right panel. This directly communicates which sectors matter and how strongly they matter.

Recommended CLI core: `--N, --mass, --coupling, --chi, --outdir`, with analysis options `--cut, --max-sectors, --sort-by`, and `--plot`.

6.2.6 CSV schema and acceptance criterion

A tidy long-form schema is

`sector,weight,entropy_contribution,rank_count,cut,model,N,m_over_g,x,chi`

The script is considered complete when it runs end-to-end at one parameter point, produces a stable CSV and one interpretable figure, and supports an honest statement of the form:

The bipartite entanglement is concentrated in a small number of constrained sectors, indicating that the physical entanglement structure is strongly organized by the gauge / symmetry constraints.

6.3 Extension B — `open_schwinger_entanglement_dynamics.py`

6.3.1 Scientific purpose This script unifies the Schwinger many-body dynamics side of the repo with the open-system infrastructure and asks how weak dissipation alters entanglement growth and tensor-network compressibility. The clean framing is:

We extend the closed-system Schwinger entanglement analysis to weakly open dynamics and track how dissipation modifies entropy growth, entanglement structure, and effective Schmidt compressibility.

6.3.2 Minimal physical scope The first version should freeze one initial state or quench, one dissipation channel, one or two small dissipation strengths, one or two representative cuts, and a short-to-moderate time window. It should always include a direct comparison between the closed case $\gamma = 0$ and one weakly open case $\gamma > 0$.

6.3.3 Dynamics equation and observable hierarchy The density matrix is assumed to satisfy a GKLS / Lindblad equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_{\mu} \left(L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{L_{\mu}^{\dagger} L_{\mu}, \rho\} \right), \quad (6.6)$$

with one chosen weak channel. If dephasing is used, one may take local operators of the form

$$L_n^{(\text{deph})} = \sqrt{\gamma} O_n, \quad (6.7)$$

where the precise on-site operator O_n must match the implementation convention and should be stated explicitly in the script header.

The script should compute:

1. **Time-dependent subsystem entropy** $S_A(t)$ or, if the evolution object is an MPO, the corresponding mixed-state reduction according to Eq. (3.13);
2. **time-dependent operator-space entanglement or Schmidt concentration** using Eq. (3.15) or cumulative retained weight at selected times;
3. **one physical many-body observable**, for example the central electric field,

$$\langle L_{\ell_c}(t) \rangle = \text{Tr}[\rho(t) L_{\ell_c}], \quad (6.8)$$

or a staggered imbalance,

$$\mathcal{J}(t) = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n \text{Tr}[\rho(t) \sigma_n^z]. \quad (6.9)$$

The inclusion of the second diagnostic is what makes the script a tensor-network / QI extension rather than a generic noise study.

6.3.4 Deliverables and figure design The planned outputs are:

- `open_schwinger_entanglement_dynamics.csv`,
- `open_schwinger_entanglement_dynamics.png`,

- `run_metadata.json`.

The recommended application-facing figure is a two- or three-panel plot with: (i) $S_A(t)$ for closed vs open dynamics, (ii) cumulative Schmidt / operator-space Schmidt weight at selected times, and (iii) one physical observable such as Eq. (6.8) or Eq. (6.9).

Recommended CLI core: `--N, --mass, --coupling, --chi, --outdir, plus --tmax, --nt, --cut, --initial-state, --quench, --gamma, --gamma-ref, --channel, and --snapshot-times`.

6.3.5 CSV schema and acceptance criterion

A tidy long-form table is

```
time, observable, value, cut, channel, gamma, model, N, m_over_g, x, chi
```

An second CSV for Schmidt or OSEE snapshots may use

```
time, rank, lambda, lambda_sq, cum_weight, cut, channel, gamma, model, N, m_over_g, x, chi
```

The script is complete when it runs for one closed and one weakly open case, produces a stable entropy-vs-time comparison, and supports an honest interpretation such as:

Weak dissipation suppresses and reshapes entanglement growth relative to the closed Schwinger quench, with corresponding changes in Schmidt-weight concentration and effective tensor-network compressibility.

6.3.6 Mathematical caution If the implementation does not have direct access to reduced density matrices of physical subsystems, the script should **not** mislabel operator-space entanglement or trajectory-averaged entropy as the subsystem entanglement of the mixed state. The observable name in the CSV, plot titles, and script docstring must follow the actual mathematical quantity being computed.

7. Theory-to-Deliverable Mapping

Theory Section	Key Equations	Repository Deliverable	Application Claim Supported
§2.2	Eqs. (2.4)–(2.10)	utils_entanglement	Change: get_schmidt_values_by_sector constrained bipartite structure
§3.1	Eqs. (3.1)–(3.2)	schwinger_entanglement	Core: entropy.py Neumann entropy observable
§3.2	Eqs. (3.3)–(3.6)	utils_entanglement	Intersector: compute_sector_resolved_entropy versus intra-sector decomposition
§3.3	Eqs. (3.7)–(3.9)	schwinger_entanglement	Symmetry: spectrum.py labeled entanglement structure
§3.4	Eqs. (3.10)–(3.12)	schmidt_decay_and_compressibility	Tensor: network compressibility
§6.2	Eqs. (6.1)–(6.5)	schwinger_symmetry_upgrade (upgrade)	Which sector actually carry the entanglement
§3.5 and §6.3	Eqs. (3.13)–(3.16), (6.6)–(6.9)	open_schwinger_entanglement_upgrade (upgrade)	How given openness modifies entanglement growth and compressibility
§4.2	Eq. (4.2)	Bond-dimension and time-step convergence checks	Numerical reliability

No core section is orphaned. The two new scripts are deliberately marked as upgrades and therefore do not invalidate the core Section 05 deliverables if they are not implemented.

8. Validation Criteria

8.1 Core ground-state checks

1. **Schmidt normalization:** $\sum_{\alpha} \lambda_{\alpha}^2 = 1$ at every cut.
2. **Entropy non-negativity:** $S(\ell) \geq 0$.
3. **Trivial Hilbert-space bound:** $S(\ell) \leq \ln \min(2^{|A|}, 2^{|B|})$.
4. **Sector-weight normalization:** $\sum_q p_q(\ell) = 1$.
5. **Entropy decomposition identity:** Eq. (3.5) must hold numerically within floating-point tolerance.
6. **Sector admissibility:** reported charge sectors must satisfy the parity-dependent bounds in Eq. (3.9) and the corresponding right-block constraints.

8.2 Additional checks for `schwinger_symmetry_resolved_entanglement.py`

7. **Sector-label consistency:** the sector label exported to CSV must match the conserved TeNPy bond charge used internally.
8. **Cut-flux interpretation check:** in the zero-background convention, the reported sector label should reproduce $L_{\ell} = q_A$ whenever that interpretation is claimed.
9. **Cumulative-weight monotonicity:** $P_{\leq k}$ in Eq. (6.2) must be nondecreasing in k and approach 1.
10. **Stable ranking:** the dominant sector weights should remain numerically stable under modest increases of χ .

8.3 Additional checks for `open_schwinger_entanglement_dynamics.py`

11. **Trace preservation:** $\text{Tr} \rho(t) = 1$ for all stored times.
12. **Hermiticity and positivity:** the evolved density matrix must remain Hermitian and positive semidefinite within numerical tolerance, or the evolution backend must guarantee these properties by construction.
13. **Closed-limit recovery:** the $\gamma = 0$ run must reproduce the trusted closed-system evolution pipeline to within solver tolerance.
14. **Observable-label honesty:** each plotted or exported entropy-like quantity must be labeled according to whether it is subsystem entropy, OSEE, or trajectory-averaged entropy.
15. **Compressibility check:** cumulative Schmidt or operator-space Schmidt weight should be reported at selected times so that tensor-network cost

claims are directly supported by data.

16. **Convergence:** the time series should remain stable under a modest increase in bond dimension, MPO bond dimension, or time-step resolution, depending on the evolution method used.

8.4 Priority criterion for stopping

Each script should be judged by whether it supports a precise application-facing sentence. If that sentence cannot be written honestly, the script should be treated as unfinished rather than padded with weak figures.

9. Notation Reference

Symbol	Meaning	Units / Range
N	Number of lattice sites	integer ≥ 4
a	Lattice spacing	length
$w = 1/(2a)$	Hopping amplitude	energy
$\mu = m$	Bare fermion mass	energy
$J = g^2 a/2$	Electric-field coupling	energy
ℓ_0	Background electric field	dimensionless; fixed to 0 here
L_n	Electric field on link n	dimensionless
ρ_n	Local staggered charge operator	$\{0, 1\}$ on even sites, $\{-1, 0\}$ on odd sites
Q, Q_A, Q_B	Total and block charges	integers in the present convention
λ_α	Schmidt values of a pure-state cut	$[0, 1]$, normalized by $\sum \lambda_\alpha^2 = 1$
ξ_α	Entanglement energies	≥ 0
$p_q(\ell)$	Sector weight at cut ℓ	$[0, 1]$, $\sum_q p_q = 1$
$S_q(\ell)$	Intra-sector entropy	nats
$h_q(\ell)$	Inter-sector Shannon contribution	nats
$c_q(\ell)$	Weighted intra-sector contribution	nats

Symbol	Meaning	Units / Range
$W(\chi')$	Cumulative retained Schmidt weight	$[0, 1]$
$\chi_{\text{eff}}(\epsilon)$	Effective bond dimension at tolerance ϵ	integer
$\rho(t)$	Time-dependent density matrix	positive, trace one
$S_A(t)$	Subsystem entropy of a mixed state	nats
$s_a(t)$	Operator-space Schmidt values of $\text{vec}(\rho(t))$	$[0, 1]$, normalized by $\sum_a s_a^2 = 1$
$S_{\text{op}}(\ell, t)$	Operator-space entanglement entropy	nats
γ	Dissipation strength in the open-system extension	≥ 0
L_μ	Lindblad jump operator	depends on chosen channel
$\mathcal{I}(t)$	Representative staggered imbalance observable	dimensionless